R FOR DATA ANALYSIS · R BASICS · LESSON 1

01 Your first calculations in R

Meet RStudio and use R as a calculator, one line at a time.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Find the four panes of RStudio
- Run a line of code with Control and Enter
- Read the answer in the console

The R code in this lesson

```
# Basic calculation
2+5
574-4
2+5/3+10*8-9
```

What you will hear (times count from the end of the intro)

Intro Welcome to your first R lesson. Take a deep breath, there is no rush. In this lesson, you will meet RStudio, write your very first lines of code, and see R give you the answers. We will go step by step, together.

0:00 This is RStudio. On the top left is the script, where we write our code. Below it is the console, where R shows the results. On the right, you see the environment, and your files.

0:15 Let's start with something simple. I write a small note first, with a hash sign: Basic Calculation. Anything after a hash is a comment. R does not run it. It is only there to help us remember.

0:32 Now our first line: two plus five. Put your cursor on the line, and press Control and Enter. On a Mac, it is Command and Enter.

0:43 Look at the console. R says seven. Wonderful, you just ran your first piece of code!

0:50 Next line: five hundred seventy four minus four. Run it the same way. The answer is five hundred seventy.

0:58 Now a longer one: two plus five divided by three, plus ten times eight, minus nine. R follows the usual math order. It divides and multiplies first, then adds and subtracts. The answer is seventy four point six six.

1:25 Notice how every result in the console starts with a one in brackets. That just means this is the first value of the answer. Nothing to worry about.

1:35 Next, I add a new heading: R Data Types. Under it, three more notes: character, numeric, and logical. These are the three kinds of values we will meet first.

2:10 Let's begin with the character type. A character is simply text, like a name. Text in R always goes inside quotation marks.

2:23 Here I type a name inside quotes, as our first example. In the next lesson, we will look at each data type closely. You did great today!